

Dang Nguyen

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Research interests

My research focuses on data-centric methods for building **efficient and reliable AI agents**. I study how to improve the training of large language and vision-language models through **synthetic data generation** and **data selection**, particularly when high-quality data is limited. More recently, I have been exploring how to enable models to **reason, plan, and act effectively** through reinforcement learning, with a focus on agent planning, orchestration, and safe tool use. Ultimately, I aim to develop agents that can learn from data and reliably make decisions and execute complex tasks over long horizons.

Education

University of California, Los Angeles

Ph.D. in Computer Science

California, USA

Sep. 2023 - Present

- Advisor: Professor Baharan Mirzasoleiman
- UCLA Graduate Dean's Scholar Award

Toyo University

B.S. in Information Networking for Innovation and Design

Tokyo, Japan

Apr. 2017 - Mar. 2021

- Toyo Top Global Scholarship A
- GPA: 4.27/4.3, Top 1/300 in the faculty
- Top thesis award

Experience

Adobe

Applied Scientist Intern

California, USA

Jun. 2026 - Sep. 2026

- Supervisors: Shubham Lohiya, Sizhe Zhou, Dr. Anay Majee
- Manager: Dr. Sajjadur Rahman
- Topic: Configurable Safety and Governance for Autonomous Agentic Systems
- Designed a runtime agent safety and governance system combining organization-level policies with a reasoning-blind bounded tool-call/result window to detect intent-action mismatches, tool-result prompt injection, and dangerous cross-tool chains; achieved 99.3% recall and 96.2% precision on 1k+ production approval requests, reducing human approval prompts by 95.5%.
- Developed a synthetic adversarial dataset generation and evaluation framework that automatically generates safety-critical agent trajectories through interactions with sandboxed backends and verifies their reachability and impact; constructed a benchmark exposing current failure modes and enabling systematic regression testing.
- Integrated the system into Adobe Experience Platform, enabling configurable runtime governance for autonomous enterprise agents.

Google Research

Student Researcher

California, USA

Sep. 2024 - Aug. 2025

- Topics: Synthetic data generation for LLMs and Data selection for LVLMS

Cisco

PhD Research Intern

California, USA

Jun. 2024 - Sep. 2024

- Supervisor: Dr. Ali Payani
- Topic: LLM Hallucination

VinAI (now Qualcomm AI)

AI Resident

Hanoi, Vietnam

Oct. 2020 - Aug. 2023

- Supervisor: Professor Nhat Ho (UT Austin)
- Topics: Optimal Transport and Model Merging
- Participated in an applied project which aims to improve the performance of object detectors in low-light conditions.
- Managed GPU resources for the VinAI Residency Program.

FPT Japan Holdings

Part-time Machine Learning Engineer

Yokohama, Japan

Oct. 2019 - Sep. 2020

- Participated in a long-term demand forecasting project for a chain pharmacy company in Japan.

Publications

(*) denotes equal contribution

1. D. Nguyen*, J. Huang*, A. Payani, B. Mirzasoleiman. Why is A+B Better Than B? A Simple Graph Perspective on Task Transfer. *ICML 2026 Workshop on Foundations of Deep Generative Models*.
2. N. Naharas*, **D. Nguyen***, N. Bulut, M. Bateni, V. Mirrokni, and B. Mirzasoleiman. Data Selection for Fine-tuning Vision Language Models via Cross Modal Alignment Trajectories. *International Conference on Machine Learning (ICML)*, 2026.
3. **D. Nguyen***, J. Li*, J. Zheng*, B. Mirzasoleiman. Do We Need All the Synthetic Data? Towards Targeted Synthetic Image Augmentation via Diffusion Models. *International Conference on Learning Representations (ICLR)*, 2026.
4. **D. Nguyen**, A. Payani, B. Mirzasoleiman. Beyond Semantic Entropy: Boosting LLM Uncertainty Quantification with Pairwise Semantic Similarity. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL Findings)* 2025.
5. **D. Nguyen***, Z. Li*, M. Bateni, V. Mirrokni, M. Razaviyayn, and B. Mirzasoleiman. Synthetic Text Generation for Training Large Language Models via Gradient Matching. *International Conference on Machine Learning (ICML)*, 2025.
6. **D. Nguyen**, W. Yang, R. Anand, Y. Yang and B. Mirzasoleiman. Mini-batch Coresets for Memory-efficient Language Model Training on Data Mixtures. *International Conference on Learning Representations (ICLR)*, 2025.
7. **D. Nguyen**, P. Haddad, E. Gan, and B. Mirzasoleiman. Changing the Training Data Distribution to Reduce Simplicity Bias Improves In-distribution Generalization. *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
8. Y. Xue, J. Siddharth, **D. Nguyen**, and B. Mirzasoleiman. Understanding the Robustness of Multi-modal Contrastive Learning to Distribution Shift. *International Conference on Learning Representations (ICLR)*, 2024.
9. K. Nguyen*, **D. Nguyen***, N. Ho. Self-Attention Amortized Distributional Projection Optimization for Sliced Wasserstein Point-Cloud Reconstruction. *International Conference on Machine Learning (ICML)*, 2023.
10. **D. Nguyen**, T. Nguyen, K. Nguyen, D. Phung, H. Bui, and N. Ho. On cross-layer alignment for model fusion of heterogeneous neural networks. *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2023. (Top 3%)
11. K. Nguyen*, **D. Nguyen***, T. A. V. Le, T. Pham, and N. Ho. Improving mini-batch optimal transport via partial transportation. *International Conference on Machine Learning (ICML)*, 2022.
12. K. Nguyen, **D. Nguyen**, Q. Nguyen, T. Pham, H. Bui, D. Phung, T. Le, and N. Ho. On transportation of mini-batches: A hierarchical approach. *International Conference on Machine Learning (ICML)*, 2022.

Honors & Awards

INTERNATIONAL

- 2026 **Amazon Fellows**, UCLA
2023 **UCLA Graduate Dean's Scholar Award**, UCLA
2017 **Toyo Top Global Scholarship A**, Toyo University
2015 **Silver medal**, 56th International Mathematical Olympiad

California, USA
California, USA
Tokyo, Japan
Chiang Mai, Thailand

DOMESTIC

- 2015 **First Prize**, Vietnam Mathematical Olympiad
2014 **Second Prize**, Vietnam Mathematical Olympiad

Hanoi, Vietnam
Hanoi, Vietnam

Professional services

- Reviewer at Conference on Neural Information Processing Systems (NeurIPS) 2022-2026
- Reviewer at the International Conference on Machine Learning (ICML) 2023-2026
- Reviewer at the International Conference on Learning Representations (ICLR) 2024-2026
- Reviewer at the International Conference on Artificial Intelligence and Statistics (AISTATS) 2023-2025
- Program Committee at AAAI 2025
- Program Committee at New Frontiers in AdvML@NeurIPS2024
- Reviewer at Workshop on Spurious Correlation and Shortcut Learning @ ICLR 2025